

Functional Analysis PhD Exam, January 2011

Do only 5 of the 8 problems.

1. State the Krein–Milman Theorem and sketch the proof.
2. Prove that if A is an element of a Banach algebra $\mathcal{U}$, then the spectrum of A is a nonempty closed subset of the closed disk in $\mathbb{C}$ with center 0 and radius $\|A\|$.
3. State the “Function Representation Theorem” involving an abelian C^* -algebra $\mathcal{U}$ in terms of $C(\mathcal{P}(\mathcal{U}))$, where $\mathcal{P}(\mathcal{U})$ is the set of all pure states of $\mathcal{U}$. Sketch the proof of this theorem.
4. State and sketch the proof of the theorem involving $\mathcal{A} \cong C(X)$, where $\mathcal{A}$ is an abelian von Neumann algebra and X is an extremely disconnected compact Hausdorff space.
5. Give the main elements of the Gelfand–Naimark–Segal construction. State the theorem involving a state ρ of a C^* -algebra $\mathcal{U}$ and a cyclic representation π_ρ of $\mathcal{U}$ on a Hilbert space H_ρ . List the main steps in the proof.
6. Let $\mathcal{U}$ be a C^* -algebra. State and prove equivalent conditions for $A \in \mathcal{U}^+$.
7. Let H be a self-adjoint operator on a Hilbert space. Define the Cayley transform $U(H)$ and show that $H \mapsto U(H)$ on the self-adjoint operators is strong-operator continuous.
8. Let $\mathcal{A}$ be the algebra of multiplications by elements of $L_\infty(S, \mathcal{S}, m)$ on $L_2 = L_2(S, \mathcal{S}, m)$. If T is a bounded operator on L_2 commuting with $\mathcal{A}$, show $T \in \mathcal{A}$. Assume $m(S) < \infty$.